

WEEK 1 PROJECT PLANNING AND STRATEGY DESIGN

AI-Based Pothole Detection and Road Condition Analysis Using Python

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Program	B.Tech – Artificial Intelligence & Data Science
Task	Week 1: Data Science Project Planning and Strategy Design

1. Project Background

Road potholes are a common road-safety and maintenance problem. They can cause vehicle damage, discomfort, traffic disruption and accidents. Manual road inspection requires time and human effort and may not provide continuous monitoring. This project proposes a Python-based data science and computer vision solution that can detect potholes from road images and support road-condition reporting.

2. Problem Statement

The main problem is to automatically identify potholes from road images and provide useful information for further inspection. The proposed system will use image data and an object-detection approach to locate potholes. The Week 1 activity focuses on planning the complete data science workflow; actual dataset collection, model training and deployment will be performed in later stages.

3. Project Objectives

- Design a complete data science workflow for pothole detection.
- Identify suitable sources and structure for road-image data.
- Plan image preprocessing, annotation, exploration and dataset splitting.
- Select a suitable object-detection approach using Python.
- Define evaluation metrics such as precision, recall and mAP.
- Plan a simple dashboard/reporting interface for future implementation.
- Identify expected challenges and mitigation strategies.

4. Project Scope

In scope: road-image collection, data preparation, pothole annotation, exploratory analysis, computer-vision model planning, evaluation planning, result visualization and future dashboard/report generation. Out of scope for Week 1: actual model training, live camera integration, production deployment and real-time road mapping.

5. Proposed Technology Stack

Area	Tool / Library	Purpose
Programming	Python	Main development language
Image Processing	OpenCV	Image loading and preprocessing
Data Handling	NumPy, Pandas	Data manipulation and analysis
Visualization	Matplotlib	Charts and exploratory analysis
Deep Learning	PyTorch	Model training framework
Object Detection	Ultralytics YOLO	Pothole detection model
Dashboard	Streamlit	Future interactive interface

6. Methodology and Strategy

Data Collection: Collect appropriately licensed road images containing potholes and normal road surfaces.

Data Cleaning: Remove corrupted, duplicate or unsuitable images and standardize image formats.

Annotation: Create bounding-box annotations around potholes using an appropriate annotation workflow.

Dataset Split: Divide data into training, validation and testing sets while maintaining useful class representation.

Preprocessing: Resize/normalize images as required and apply suitable augmentation during training.

Model Development: Train and compare an object-detection model using the planned Python/YOLO pipeline.

Evaluation: Measure precision, recall, mAP and inference time on validation/test data.

Visualization & Reporting: Display detected potholes and prepare summary reports for future use.

7. Planned System Architecture

Input Road Image → Image Preprocessing → Pothole Detection Model → Bounding Box & Confidence → Result Analysis → Visualization/Dashboard → Report Generation

8. Evaluation Plan

- Precision: proportion of detected potholes that are correct detections.
- Recall: proportion of actual potholes successfully detected.
- mAP@0.5: overall object-detection performance measure.
- Inference Time: time required to process an image.
- Qualitative Review: visual inspection of detection results and false positives.

9. 35-Hour Project Timeline

Stage	Planned Work	Hours
Planning & Problem Understanding	Problem definition, objectives and scope	4
Data Strategy	Data sources, format, annotation and splitting plan	5
Data Preparation Plan	Cleaning, preprocessing and augmentation strategy	5
Exploratory Analysis Plan	EDA, distributions and visualization design	4
Model Strategy	YOLO/object-detection approach and experiment plan	6
Evaluation Strategy	Metrics, test procedure and error analysis	4
Dashboard & Reporting Plan	Visualization, reporting and future deployment design	3
Documentation & Review	Diagrams, final documentation and quality review	4

Total Planned Effort: 35 hours

10. Expected Outcomes

- A clearly documented data science project architecture.
- A structured plan for dataset preparation and model development.
- Defined evaluation metrics and testing methodology.
- A future prototype capable of detecting potholes in road images.
- A foundation for visualization and road-condition reporting.

11. Potential Challenges and Mitigation

Challenge	Mitigation
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Uneven image quality	Use preprocessing and augmentation; maintain diverse samples.
Insufficient or biased data	Collect varied road conditions, locations and lighting conditions.
False detections	Review errors, improve annotations and tune confidence thresholds.
Overfitting	Use validation data, augmentation and controlled experimentation.
Deployment limitations	Measure inference time and plan lightweight deployment options.

12. Conclusion

This Week 1 plan establishes the strategy for an AI-based pothole detection project using Python. The focus is on understanding the problem, defining scope, planning the data pipeline, selecting appropriate tools, designing the model and evaluation strategy, and allocating project time. The plan will serve as the foundation for the technical implementation in later stages.

13. References / Planned Resources

- Python documentation and standard data-science libraries.
- OpenCV documentation for computer vision.
- PyTorch documentation for deep learning.
- Ultralytics documentation for YOLO-based object detection.
- Streamlit documentation for future dashboard development.